Question 2 Report: Reduced Reproduction of Disentangled Representation Learning for Environment-agnostic Speaker Recognition

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March 23, 2026

1 Task Summary

This report addresses Question 2(b)-(d):

1. Implement the method from *Disentangled Representation Learning for Environment-agnostic Speaker Recognition* (arXiv:2406.14559), or a justified reduced reproduction, and compare against a suitable baseline.
2. Report results with appropriate metrics and visualizations.
3. Propose one critique-motivated improvement and evaluate whether it improves performance.

The implementation in this submission is a **reduced reproduction**. It preserves the core idea of disentangling speaker and environment factors with reconstruction and speaker supervision, while simplifying backbone complexity, dataset scale, and objective design compared to the full paper.

2 Paper Motivation and Method (Brief)

The paper proposes learning speaker embeddings that are robust to environmental variability by disentangling a base embedding into:

- speaker-related information, and
- residual/environment-related information.

An auto-encoder style disentangler and multiple objectives are used to make the speaker code discriminative while suppressing nuisance environmental factors.

3 Implementation in This Work

3.1 Data and Features

- Dataset root: `data/LibriSpeech/dev-clean` (configured in `configs/config.yaml`).
- Input feature: 80-bin log Mel filterbank extracted from 16 kHz audio.
- Sequence handling: each utterance is truncated/padded to 300 frames.

3.2 Baseline System

The baseline follows a lightweight disentanglement pipeline:

- **Feature extractor** (`models/feature_extractor.py`): two 1D convolution layers + global average pooling + linear projection to 192-D embedding.
- **Disentangler** (`models/disentangler.py`): MLP encoder mapping 192 → 256, then two heads:
  - speaker code $z_{spk} \in R^{128}$,
  - environment code $z_{env} \in R^{128}$.
- Decoder reconstructs embedding from $[z_{spk}, z_{env}]$.
- Speaker classifier predicts speaker ID from z_{spk}.

Baseline training objective:

$$\mathcal{L}_{base} = \underbrace{\|\hat{e} - e\|_2^2}_{\mathcal{L}_{recon}} + \underbrace{\mathrm{CE}(\hat{y}, y)}_{\mathcal{L}_{cls}}. \tag{1}$$

3.3 Critique-Motivated Improvement

Critique: The baseline only applies speaker classification to z_{spk}. This does not explicitly enforce pairwise metric structure in embedding space for verification, which is the downstream task.

Improvement: Add supervised contrastive loss on z_{spk} (implemented in `improved_exp.py`) to pull same-speaker samples together and push different-speaker samples apart.

Improved objective:

$$\mathcal{L}_{improved} = \mathcal{L}_{recon} + \mathcal{L}_{cls} + 0.5\,\mathcal{L}_{contrastive}. \tag{2}$$

This is consistent with the paper's disentanglement spirit while directly optimizing a verification-friendly representation geometry.

3.4 Training and Evaluation Protocol

- Split: 80/20 train/test, seeded with 42, and persisted via `results/split_indices.pt` for reproducible evaluation.
- Optimizer: Adam, learning rate 10^{-4}, batch size 32, 50 epochs.
- Device: CUDA (per config).
- Verification scoring: dot-product similarity between speaker codes on all test pairs.
- Operating point: threshold at EER.

4 Results

4.1 Quantitative Comparison

Metrics are taken from `results/eval_results.txt` (baseline) and `results/improved_results.txt` (improved).

Table 1: Baseline vs improved model performance on the held-out test split.

Metric	Baseline	Improved	Change
EER (lower better)	0.2833	0.1791	-0.1042
ROC-AUC (higher better)	0.7966	0.9093	+0.1127
PR-AUC (higher better)	0.1119	0.2625	+0.1505
Accuracy @ EER threshold	0.7167	0.8208	+0.1041
Precision @ EER threshold	0.0624	0.1075	+0.0451
Recall @ EER threshold	0.7167	0.8210	+0.1042
F1 @ EER threshold	0.1148	0.1901	+0.0754

Relative effects:

- EER decreases by about 36.8% relative.
- ROC-AUC improves by about 14.1% relative.
- PR-AUC improves by about 134.5% relative.

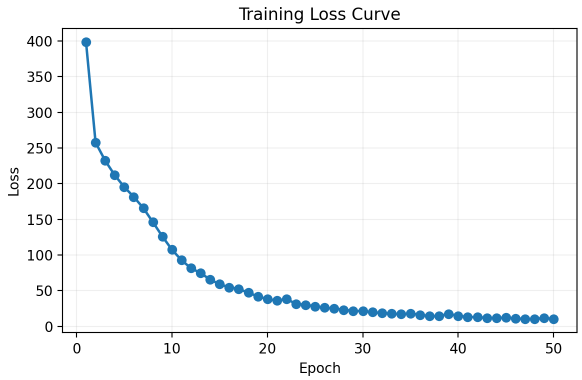
4.2 Confusion Matrices at EER Threshold

Table 2: Confusion matrix entries (test pairs).

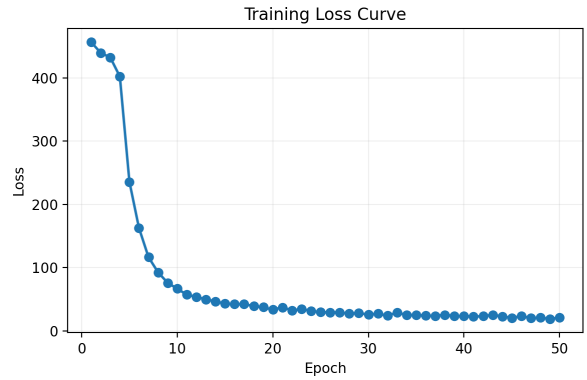
Model	TN	FP	FN	TP
Baseline	102011	40317	1060	2682
Improved	116829	25499	670	3072

The improved model simultaneously reduces both false positives and false negatives, which explains gains in EER, ROC-AUC, and F1.

4.3 Visualizations

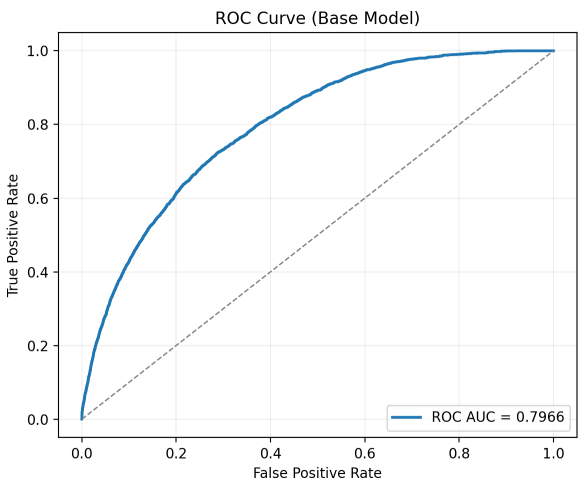


(a) Baseline training loss

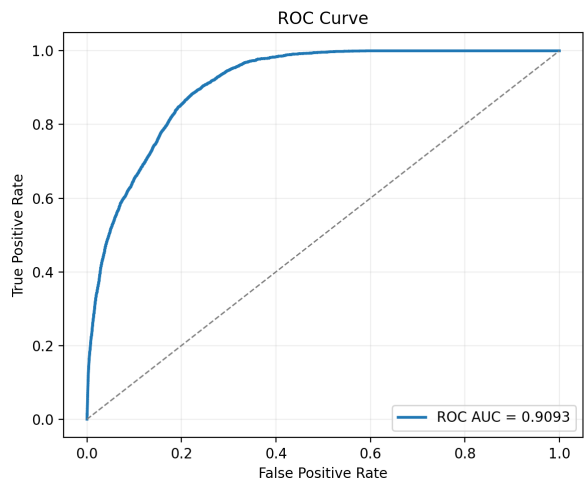


(b) Improved training loss

Figure 1: Training loss curves.

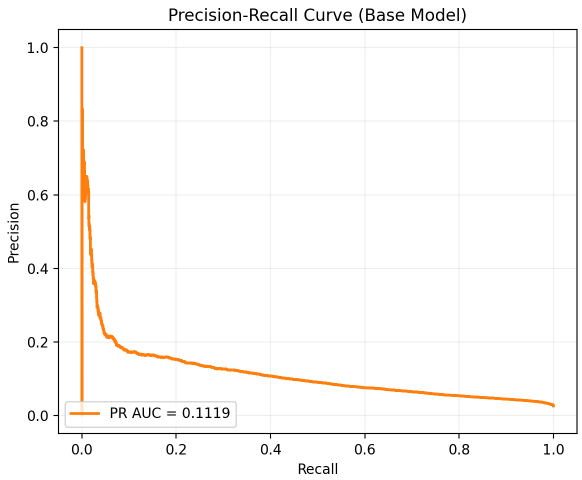


(a) Baseline ROC

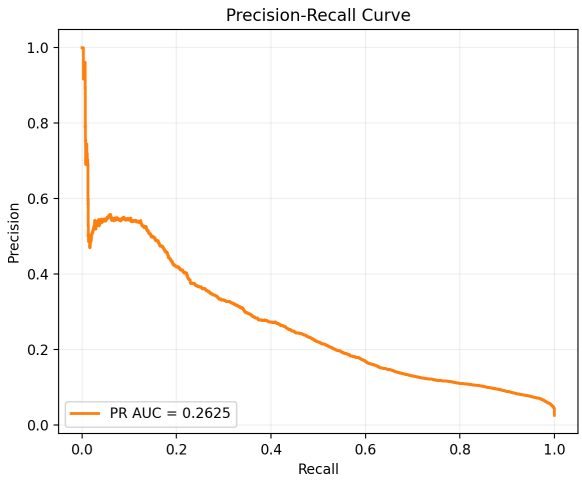


(b) Improved ROC

Figure 2: ROC comparison.

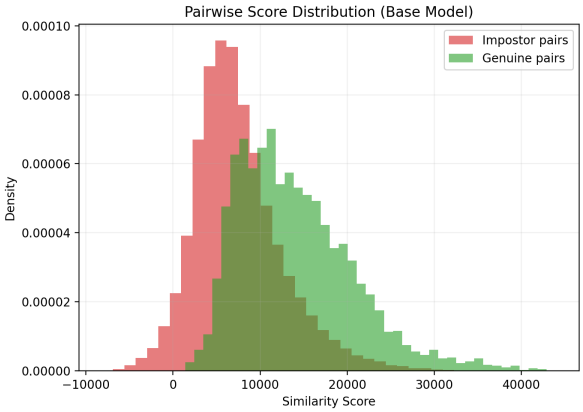


(a) Baseline PR

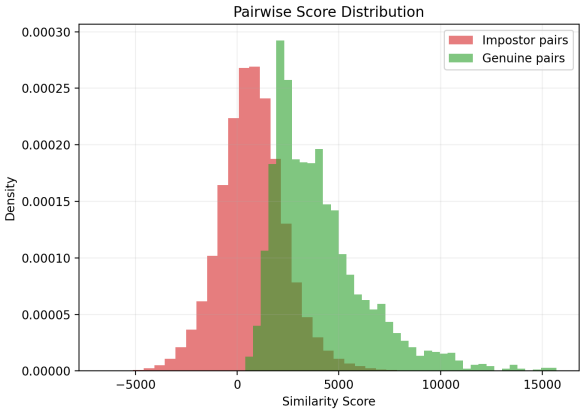


(b) Improved PR

Figure 3: Precision-recall comparison.

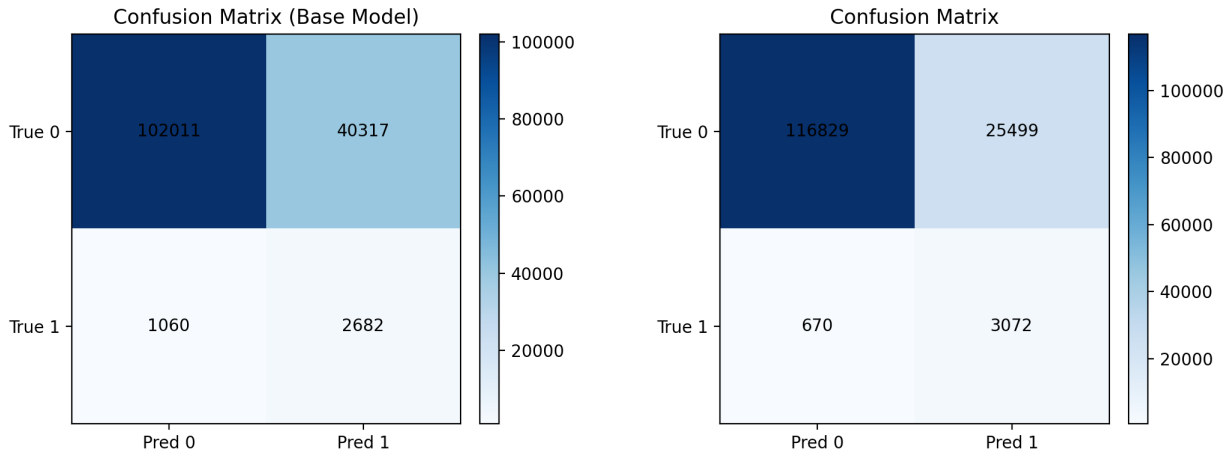


(a) Baseline score distributions



(b) Improved score distributions

Figure 4: Pairwise similarity score distributions for impostor and genuine pairs.



(a) Baseline confusion matrix

(b) Improved confusion matrix

Figure 5: Classification behavior at EER operating point.

5 Discussion

5.1 Why This Is a Justified Reduced Reproduction

Compared to the full paper, this implementation simplifies:

- backbone extractor complexity,
- objective set (fewer disentanglement constraints), and
- scale/benchmark breadth.

However, it faithfully captures the core mechanism: disentangling speaker and residual factors with reconstruction plus speaker discrimination, then evaluating verification behavior under pairwise scoring.

5.2 Improvement Evaluation

The proposed improvement (contrastive speaker-code supervision) yields consistent gains across threshold-free metrics (ROC-AUC, PR-AUC), operating-point metrics (EER, F1), and confusion matrix counts. This supports the claim that explicit metric-structure supervision makes the disentangled speaker code more suitable for verification.

6 Reproducibility and Deliverable Mapping

This submission includes:

- `train.py`: baseline training.
- `eval.py`: baseline evaluation on held-out split.
- `improved_exp.py`: improved training + evaluation with contrastive term.
- `configs/config.yaml`: experiment configuration.

- `results/`: checkpoints, metric files, and plots used in this report.
- `q2_readme.md`: commands and checkpoint provenance.

Key checkpoints:

- Baseline: `results/model.pth` with metrics in `results/eval_results.txt`s.
- Improved: `results/improved_model.pth` with metrics in `results/improved_results.txt`.

7 Conclusion

A reduced reproduction of environment-agnostic disentangled speaker representation learning was implemented and compared against a suitable baseline. The critique-motivated contrastive enhancement improved all reported verification metrics, with substantial gains in EER, ROC-AUC, and PR-AUC. The results indicate that strengthening speaker-code geometry is an effective extension to the basic disentanglement objective in this setup.

Reference K. Nam, H.-S. Heo, J.-w. Jung, and J. S. Chung, "Disentangled Representation Learning for Environment-agnostic Speaker Recognition," arXiv:2406.14559, 2024. URL: `https://arxiv.org/abs/2406.14559`